Bluetooth Classic 'Message Access Profile'

How to obtain templates for correct MAP OBEX objects

The Problem

Bluetooth SIG provides several specifications, including:

- MAP_v1.4.2.pdf
- GOEP_v2.1.1.pdf
- Test specification (MAP.TS.p14.pdf)
- Authoritative test tool PTS (v8.6.x currently)

These documents provide the necessary definitions for OBEX body objects (vCard, XML, etc.) in BNF and/or DTD form, which are essential for creating compliant applications or devices with Bluetooth Classic Message Access Profile v1.4.2.

However, creating these OBEX objects to pass all relevant PTS test cases is often a cumbersome and error-prone trial-and-error process. Feedback from PTS is typically minimal, and the tool tends to hang or crash, adding to the challenge.

The Solution

After investigation, it was discovered that PTS compares the body-objects of incoming messages with templates stored in its working directory. These templates can be found in the following directory:

`C:\Users\xyz\AppData\Roaming\Bluetooth SIG\Bluetooth PTS 8.6.0.6\install\5C26A5D\Data\Others\OBEX\MAP`

With this discovery, the implementation and test cycle typically follows these steps:

1. Select the Next Test Case in PTS

   Example: "MAP/MSE/MMN/BV-11-C"

2. Identify the Expected Body Object Type in the Description

   Example: 'ParticipantChatStateChanged' in the description "Verify that the MSE sends a 'ParticipantChatStateChanged' event-report when the chat-state of a participant is changed."

3. Search the PTS template directory

   Example: Search for "Chat" in the directory `C:\Users\xyz\AppData\Roaming\Bluetooth SIG\Bluetooth PTS 8.6.0.6\install\5C26A5D\Data\Others\OBEX\MAP\put\ParticipantChatStateChanged_event_report_1_2`

```
<MAP-event-report version = "1.2">
<event type = "ParticipantChatStateChanged" sender_name="PTS" conversation_id="3909231965"
last_activity="20160625T133700" chat_state="1" participant_uci="skype:beastmode2"/>
</MAP-event-report>
```

4. Use the File's Contents as the Template

   This file's contents can now be used as the template for the OBEX body-object required for being compliant with the specs and passing the PTS test cases.